C4.3 How do bonding and structure affect properties of materials?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*	*Linked learning opportunities*
Different materials can be made from the same atoms but have different properties if they have different types of bonding or structures. Chemists use ideas about bonding and structure when they predict the properties of a new material or when they are researching how an existing material can be adapted to enhance its properties.	1. explain how the bulk properties of materials (including strength, melting point, electrical and thermal conductivity, brittleness, flexibility, hardness and ease of reshaping) are related to the different types of bonds they contain, their bond strengths in relation to intermolecular forces and the ways in which their bonds are arranged, recognising that the atoms themselves do not have these properties	**Specification links** • ionic bonding and structure (C2.3) • metallic bonding (C3.1) • covalent bonds and intermolecular forces (C3.4) **Practical work:** • Testing properties of simple covalent compounds, giant ionic and giant covalent substances, metals and polymers.
Carbon is an unusual element because it can form chains and rings with itself. This leads to a vast array of natural and synthetic compounds of carbon with a very wide range of properties and uses. 'Families' of carbon compounds are homologous series.	2. recall that carbon can form four covalent bonds 3. explain that the vast array of natural and synthetic organic compounds occurs due to the ability of carbon to form families of similar compounds, chains and rings	**Specification link:** • The alkanes as a homologous series. (C3.4) **Ideas about Science:** • Identify patterns in data related to polymers and allotropes of carbon.(IaS2) • Use and limitations of a model to represent the structures of a range of materials. (IaS3)

C4.3 How do bonding and structure affect properties of materials?		
Teaching and learning narrative	**Assessable learning outcomes** *Learners will be required to:*	***Linked learning opportunities***
Polymer molecules have the same strong covalent bonding as simple molecular compounds, but there are more intermolecular forces between the molecules due to their length. The strength of the intermolecular forces affects the properties of the solid. Giant covalent structures contain many atoms bonded together in a three-dimensional arrangement by covalent bonds. The ability of carbon to bond with itself gives rise to a variety of materials which have different giant covalent structures of carbon atoms. These are allotropes, and include diamond and graphite. These materials have different properties which arise from their different structures.	4. describe the nature and arrangement of chemical bonds in polymers with reference to their properties including strength, flexibility or stiffness, hardness and melting point of the solid 5. describe the nature and arrangement of chemical bonds in giant covalent structures 6. explain the properties of diamond and graphite in terms of their structures and bonding, include melting point, hardness and (for graphite) conductivity and lubricating action 7. represent three dimensional shapes in two dimensions and vice versa when looking at chemical structures e.g. allotropes of carbon M5b 8. describe and compare the nature and arrangement of chemical bonds in ionic compounds, simple molecules, giant covalent structures, polymers and metals	